

Arslan Kenbayev

+82 10 7921 7883 | kenbaevarslan@gmail.com | linkedin.com/in/arslan-kenbayev | github.com/arskenchik | arskenchik.github.io

EDUCATION

KAIST

B.S. Electrical Engineering

National School of Physics and Math

Math and Physics, GPA: 3.95/4.00, Red Diploma

Aug. 2022 – Feb. 2027

Daejeon, South Korea

Sep. 2018 – May 2022

Almaty, Kazakhstan

EXPERIENCE

Research Assistant

QAIST

Aug. 2025 – Present

Almaty, Kazakhstan

- Advancing wildfire detection by developing sub-mW neuromorphic IoT sensor nodes for field-deployed wireless mesh networks.
- Reduced integration risk by writing system-level electrical interface specs and coordinating hardware/software requirements with KAIST researchers and industry partners.

IoT Device Engineer Intern

AXInvent

May 2025 – Mar. 2026

Daejeon, South Korea

- Accelerated smart agriculture prototyping by selecting components, designing I2C/SPI/UART sensor schematics and power rails, and assembling analog/digital circuits on universal PCB.
- Validated field IoT sensor data integrity through oscilloscope/DMM bring-up, sensor debugging, and edge-to-cloud integration across Arduino, Jetson, Raspberry Pi, and server via gRPC/HTTP/2.

Semiconductor Engineer Intern

KAIST IOEL Lab

May 2025 – Aug. 2025

Daejeon, South Korea

- Achieved electroluminescence in organic OLEDs by fabricating CO₂-derived/organosilicon thin-film stacks and characterizing performance through forward-bias I-V sweeps.
- Improved OLED optimization workflow by tuning layer composition from I-V data toward lower turn-on voltage and higher luminance for healthcare organic electronics.

Automation Engineer Intern

AAEngineering

July 2024 – Sep. 2024

Almaty, Kazakhstan

- Improved gold ore segregation readiness by validating sensor integration in SolidWorks, analyzing multi-node RF/IoT topology, and prototyping automated control logic.

Software Engineer Intern

KAIST CIS Lab

Feb. 2024 – June 2024

Daejeon, South Korea

- Extended KakaoMap route planning by implementing the CCH algorithm in C++ to improve dynamic road-network query performance under Prof. Heejin Ahn.

PROJECTS

Pipelined RISC-V CPU | *System Verilog, Verilog, ModelSim* | [\[github\]](#)

Sep. – Nov. 2025

- Verified 100% testbench pass rate for a synthesizable 5-stage RISC-V pipeline CPU by implementing hazard detection/forwarding in SystemVerilog and debugging ModelSim waveform timing faults across IF/ID/EX/MEM/WB stages.

Ultra-Low-Power Smartwatch Firmware | *C, Arduino Nano 33 BLE, nRF5 SDK* | [\[github\]](#)

Mar. – June 2025

- Reduced smartwatch average current draw to 85 μ A (DMM-verified) by implementing IMU duty cycling, OLED auto-sleep, and coordinated I2C sensor polling across LSM6DS step counter, APDS-9960 gesture sensor, and SSD1306 display, with BLE GATT server firmware on nRF52840 for client synchronization.

Capstone: Industrial Anomaly Detection | *Python, DCASE 2020, Git, Linux*

Mar. – June 2025

- Enabled mechanical failure detection under real-world noise by training an unsupervised autoencoder on ToyADMOS/MIMII acoustic datasets and applying reconstruction error thresholding.

Digital Clock | *Verilog, EDA Playground*

Feb. – June 2024

- Verified digital clock functionality through signal waveform analysis by designing FSMs, flip-flops, counters, and MUXes in System Verilog/Verilog.

TECHNICAL SKILLS

Hardware & Embedded: PCB Design, Soldering, Firmware Dev (nRF5 SDK, Segger Studio), Sensor Integration, I2C/SPI/UART/BLE, I-V Characterization, Oscilloscope/DMM

Languages: SystemVerilog, Verilog, C, C++, Python, MATLAB, R, Java

Platforms: Arduino Nano 33 BLE, Nvidia Jetson, Raspberry Pi, RISC-V

EDA / CAD: Fusion 360, SolidWorks, ModelSim, EDA Playground

Developer Tools: Git, VS Code, Linux